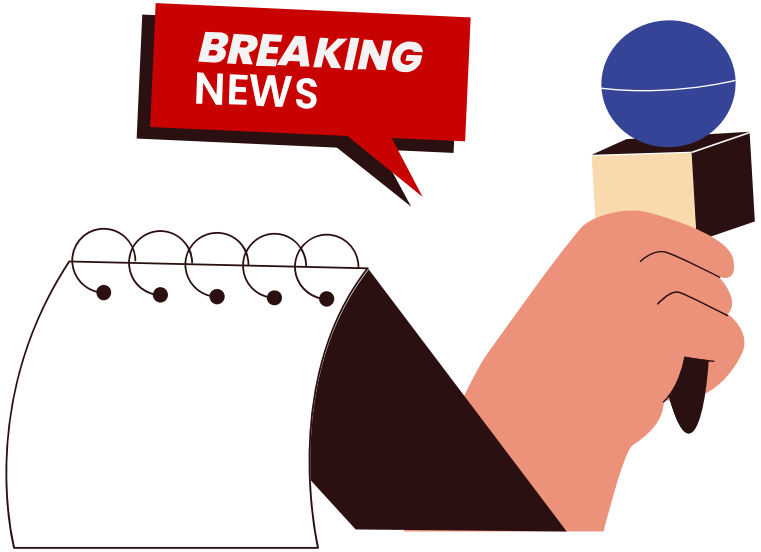
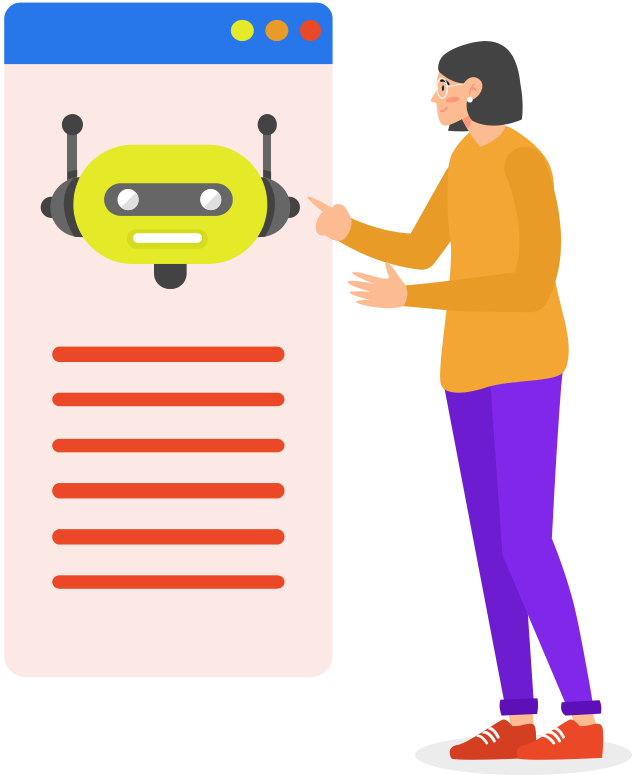
Final Project
AI across companies and industries



Next-Gen NLP: Transformers, LLMs and AgenticAI

Mateo Galvis López

Agenda



1. Project description and main goal
2. Article clean-up and filtering
3. Topic detection
4. Entity extraction (organizations, technologies)
5. Topic-level sentiment analysis
6. Entity-level sentiment analysis
7. Visualization of sentiment analysis over time

Project description and main goal

General objectives

- Identify industries and their companies in more risk to be affect by AI.
- Impact positive, negative about the industries and the technologies.
- Recommendations about AI, of successful and unsuccessful cases.

Article clean-up and filtering

Initial dataset

Information **199,898 articles with 5 columns**

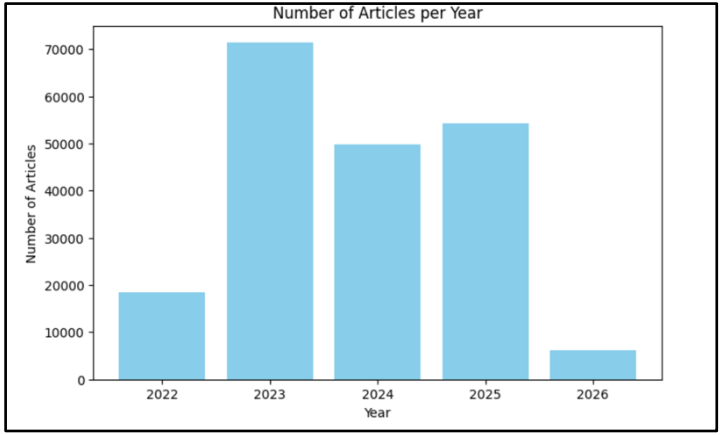
- URL: Direction of the article in the web.
- Date: The date of the publication of the article.
- Language: the language of the article.
- Title: Name of the article.
- Text: Main content of the article.

Initial cleaning steps and additional features

- Drop the column of language since all the values are equal. All articles are in English.
- Secure dtypes of the columns. For instance, the date must be in datetime format.
- Created additional columns based on the date, such as year and month.
- Create a two columns that counts the number of words in the title and text by article.

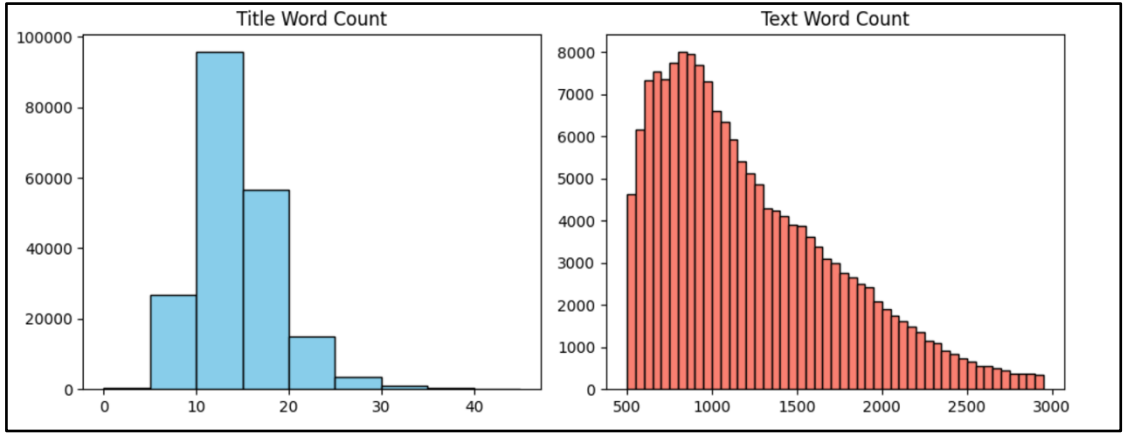
Article clean-up and filtering

Explore distribution of articles by year



2023 has the highest concentration of articles with 36%, followed by 2025 and 2024 with an average frequency of around 25%. In addition, in 2022, the number of AI articles was not a main trend topic in contrast with the next years, and the information available from 2026 only contains data until February.

Number of words



- Title word count: min 4, avg 14, and max 1000, no missing titles
- Text word count: min 4, avg 1231, and max 4000, some articles have little content.
- In almost all the text, the first sentence is the title.
- Some articles are not related to AI in companies, they include information on AI image generation, AI videos, investments in crypto bonkers, among others.

Article clean-up and filtering

1 Using regex, we remove the main source title to avoid bias in future steps.

Before:

- Sample title 1: 4 Principles to Help Federal Agencies Adopt AI Ethically and Securely **| FedTech Magazine**
- Sample title 2: OpenAI: OpenAI working with ex-Twitter India head to navigate AI landscape**- The Economic Times**
- Sample 3: Orca AI Secures $72.5 Million Investment to Scale Autonomous Shipping Solutions

After:

- Sample title 1: 4 Principles to Help Federal Agencies Adopt AI Ethically and Securely
- Sample title 2: OpenAI: OpenAI working with ex-Twitter India head to navigate AI landscape: report Sample 3: Orca
- Sample 3: AI Secures $72.5 Million Investment to Scale Autonomous Shipping Solutions

2 Remove unnecessary tokens: multiple white spaces, multiple jumps of lines, standardize apostrophes.

Before:

“ Skip to main content \n \n \n Skip to search \n AI in can’t protect privacy and 'real is learning to stole people work‘ …”

After:

“Skip to main content \nSkip to search\nAI in can't protect privacy and 'real is learning to stole people work' …”

3 Remove by default the first line of the text that have the original unclean title.

Article clean-up and filtering

4 Remove content related to other articles, sources, and videos using key words after the main content of the document

phrases = [r"\bmore stories\b", r"\bmore on\b", r"\bmore from Star Tribune\b", r"\brelated topics\b", r"\blatest shows\b", r"\bpost navigation\b", r"\brelated articles\b", r"\bread next\b", r"\bread more\b", r"\bread more news\b", r'\bview comments recommended\b', r'\bsubscribe to\b', r'\brecommended stories\b',...]

All the content after this keywords was removed

5 Since the unclean title of the article was removed from the first line of text, it was researching the next appearance of the title in the article. Therefore, when it was found, we removed all the above information related to the webpage obtained using web scraping: menu options, email, publicity. This represents a huge improvement in considering only the main content.

Example:

Information after title

Insider Sign up today to receive premium content!
Sign Up MENU
Tips & Tactics Video CapITal Events More

Title and main content

Jan 16 2025
4 Principles to Help Federal Agencies Adopt AI Ethically and Securely
by Tanya Candia
Federal civilian agencies understand that artificial intelligence will....

Information before content

More on
Data-Driven Decisions
Innovation

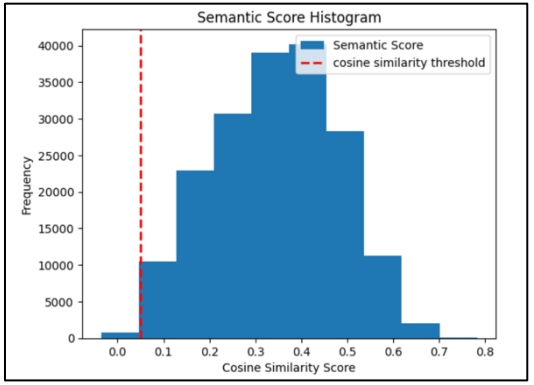
Article clean-up and filtering

6 Remove articles with fewer than 50 words, focus on video and image generation. In total, 6.98% records were eliminated, which represents 14k.

7 Combine title_clean + text_clean in one column

8 Develop a reference sentence for a potential filter article not related to the main topic. The transformer model called **all-MiniLM-L6-v2** was used to embed the reference sentence and the first 500 tokens of each article, then compute cosine similarity, and filter documents using a threshold.

reference_sentences = [
"AI protections and regulations for companies",
"The role of Prompt Engineering",
"Industry applications of artificial intelligence",
"The impact of AI on the workforce and job market".....]



We use a threshold of 5%

In conclusion:

- The initial **average words in the text** column **was 1231 words**, after cleaning it was **reduced to 757**, which is a **decrease of 38%** on the average number of words per article.
- In total, **186,028 records are considered**.

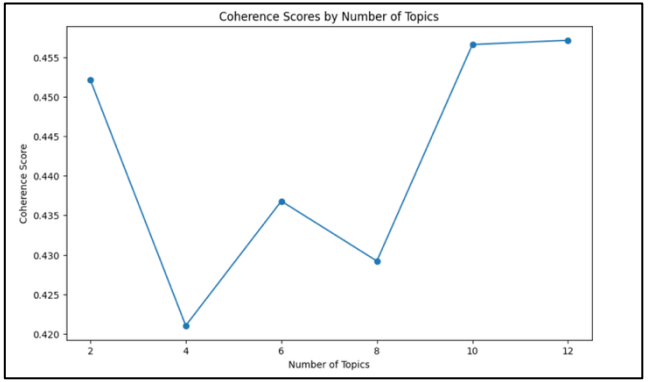
Topic detection

To begin understanding the general opinion in articles on the impact of AI on the job market, we applied a topic modeling method. A classic method, Latent Dirichlet Allocation (LDA), was selected to discover the hidden topics in the articles.

1. The first step was to use a lemmatization process to normalize the text and improve comparisons between articles. So, the key steps were lowercase tokens, removing stop words, deleting punctuation, and not considering tokens shorter than 1 character.

2. With the normal text, a bigram approach was applied to common concepts that are written together, such as artificial intelligence.

3. It was employing a grid search to establish the right number of topics based on the Coherence Scores from 2 to 12, which increase by a step of 2.



The Coherence graph shows that after 2, the values start increasing, we can see that between 10 and 12, it reached a stable score. Therefore, we chose **10 as our number of topics** with a score of 0.4571. In addition, the distance map choose **clear separate between topics**; there is only some overlap between two topics.

Topic detection

So, the 10 topics have the following keywords:

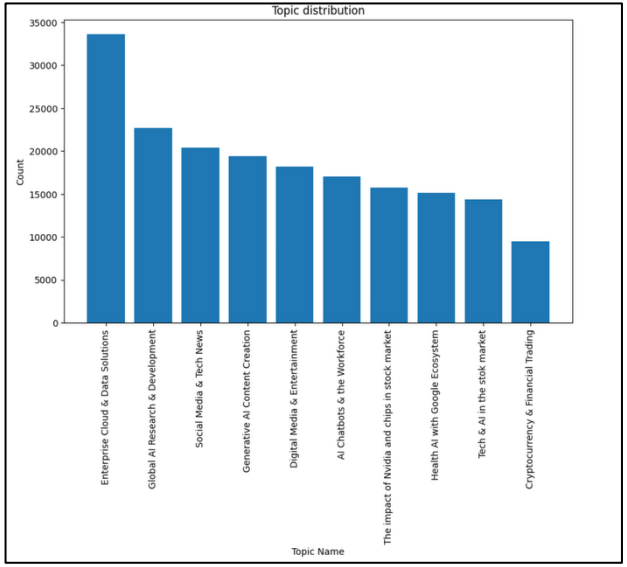
Topic	Top 7 words	Name of the topic
1	google, health, patient, help, artificial_intelligence ,image, google, learning	Health AI with Google Ecosystem
2	Chatbot, chatgpt, pay, salary, people, time, work	AI Chatbots & the Workforce
3	share, market, nvidia, chip, industry, stock, generative	The impact chips in the stock market
4	video, game, time, service, business, experience, product	Future Digital entertainment & gaming
5	Story, email, report, Microsoft, twitter, openai	Social Media & Tech News
6	Image, content, create, generative_ai, platform, tool, user	Generative AI Content Creation
7	stock, market, tech, price, million, artificial_intelligence, share	Tech & AI in the stock market
8	artificial_intelligence, research, innovation, global, development, risk, lead	Global AI Research & Development
9	customer, cloud, data, power, provide, solution, datum	Enterprise Cloud & Data Solutions
10	trading, time,, crypto, rate, price, trend, investment	Cryptocurrency & Financial Trading

- According to the LDA method and the keywords/bigrams of the topics, the industries most affected by AI are tech, research, finance, health, social media, cloud solutions, crypto, content creation, and gaming. These industries are having a strong transformation.
- It appears across several topics' top tech companies: OpenAI, Google, Nvidia, Microsoft, Apple, etc. This suggests that the ”Big Five” play a key role across several industries, not only tech.
- The terms salary, price, people, and work are common in several topics, the public is aware that AI is changing the job market fast.

Topic detection

Distribution of topics across articles

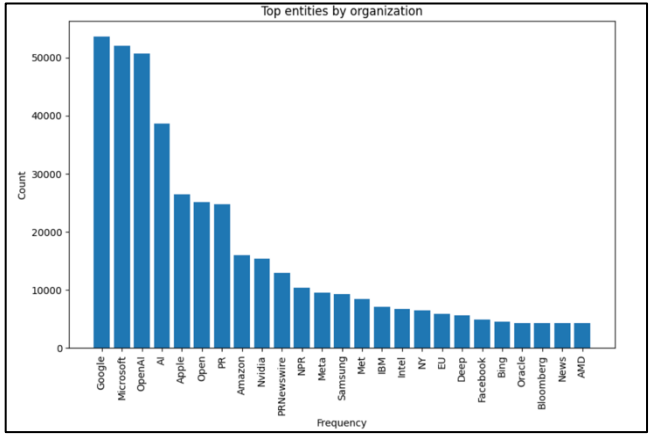
Topic Name	Frequency	Percentage
Enterprise Cloud & Data Solutions	33617	18%
Global AI Research & Development	22677	12%
Social Media & Tech News	20443	11%
Generative AI Content Creation	19400	10%
Digital Media & Entertainment	18186	10%
AI Chatbots & the Workforce	17018	9%
The impact chips in the stock market	15738	8%
Health AI with Google Ecosystem	15101	8%
Tech & AI in the stok market	14384	8%
Cryptocurrency & Financial Trading	9465	5%

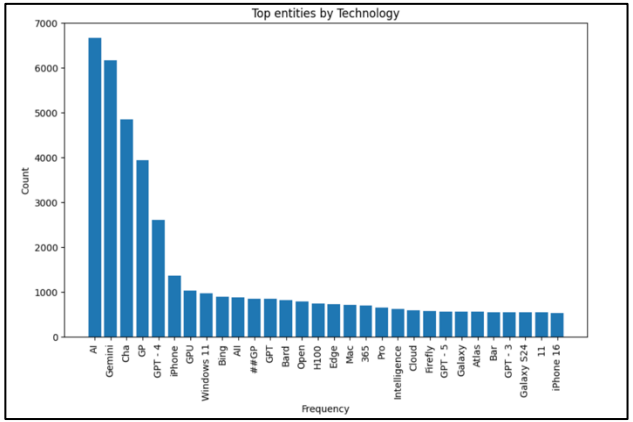


- The topic with the greatest number of articles is “Cloud and & Data solution” with 18% of the total records, the rest of the topics present a similar distribution between 8%-12% of the total news about AI. However, the last topic, Cryptocurrency & Financial Trading, has the lowest concentration with 5%.

Entity extraction (organizations, technologies)

To obtain the organizations and technologies entities in the articles, we applied a (NER) Bert-base-cased pretrain model of Hugging Face called ***ner-bert-base-cased-ontonotesv5-englishv4***. It was fine-tuning using different types of entities, such as organization, product, event, and person on several English datasets around weblogs, pr newsire, newsgroups, among others.





- For the top entities by organization, we can see that tech companies are very present such as: Google, Microsoft, OpenAI, Apple, Amazon, Meta, Samsung represent the highest frequency. In addition, appears semiconductors companies (AMD), financial institutions (Bloomberg) , database management (Oracle) and manufactures of electricity products (Samsung) .

- In the technology's entities, it was found a strong presence of LLMs (Encoder models), such as Gemini, Bard, GPT-5, GPT-4, GPT-3 in contrast with more traditional tools Bing, Edge, Microsoft 365, Windows. Moreover, a lot of cellphone and electronics products exhibited a considerable frequency iPhones, Galaxy, Mac.

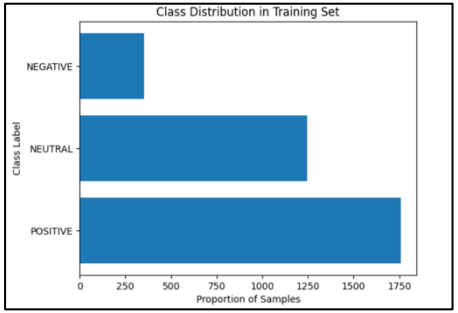
Sentimental analysis training model

•For the sentimental analysis section, it was a requirement to train a model, rather than taking an algorithm already designed for this specific use case.
•A basic Bert model was selected to be fine-tuned using a sentiment dataset. Therefore, a **distilbert-base-uncased** model was chosen, since it is a small, faster, lighter version compared to standard Bert language model and retains on average 97% of its capabilities.
•The dataset selected to perform the tuning process is **community-datasets/per_sent**, which captures the sentiment of a writer in a news article. It contains 5.3k documents with the possible labels (Positive, Neutral, Negative). For this use case, a truncation was applied in the tokenization of the articles.

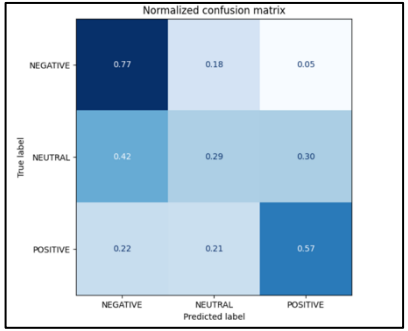
Partition

Partition	Documents
Train	3355
Validation	827
Test	579

Distribution of classes



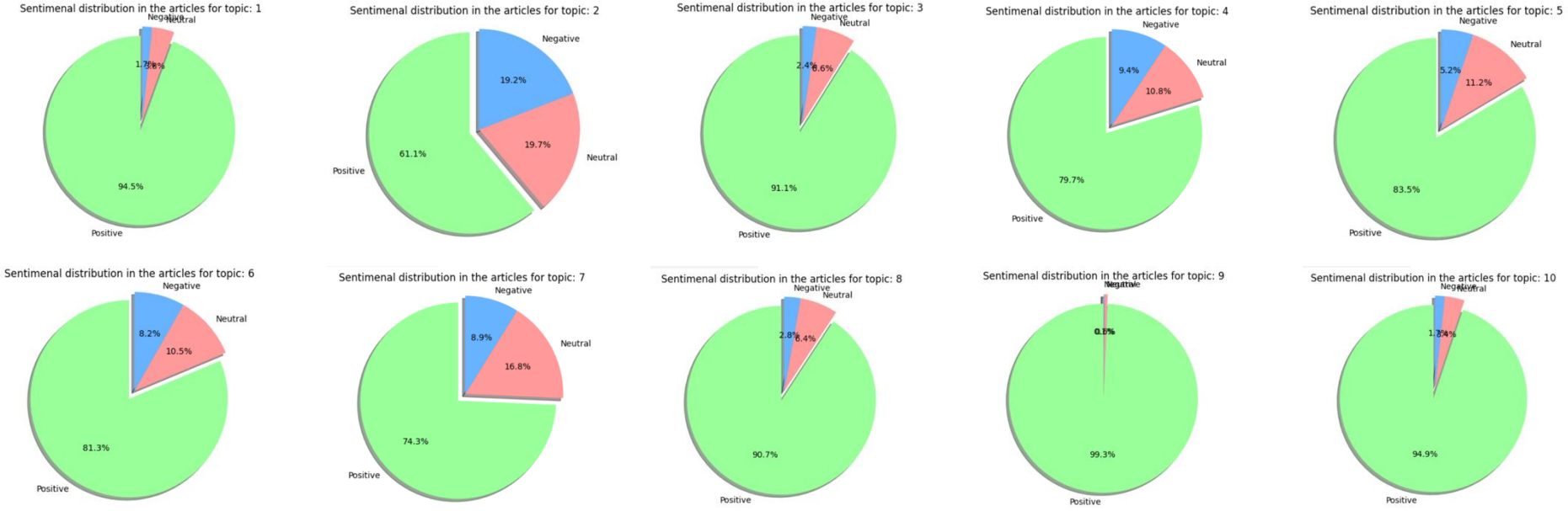
Confusing matrix Test



- For distribution of classes, it is clear that there is an imbalance issue, the negative class only have 10% (335) of the records, in contrast with neutral 37%(1242) and positive with 52% (1745). Consequently, it was necessary to use a weight parameter, the inverse of the frequency, in the tuning process to control this aspect.
- The confusing matrix displays the actual vs predicted in the test sample. We can see that the model becomes more confused about the neutral and positive classes than about the negative class.

Sentimental analysis training model

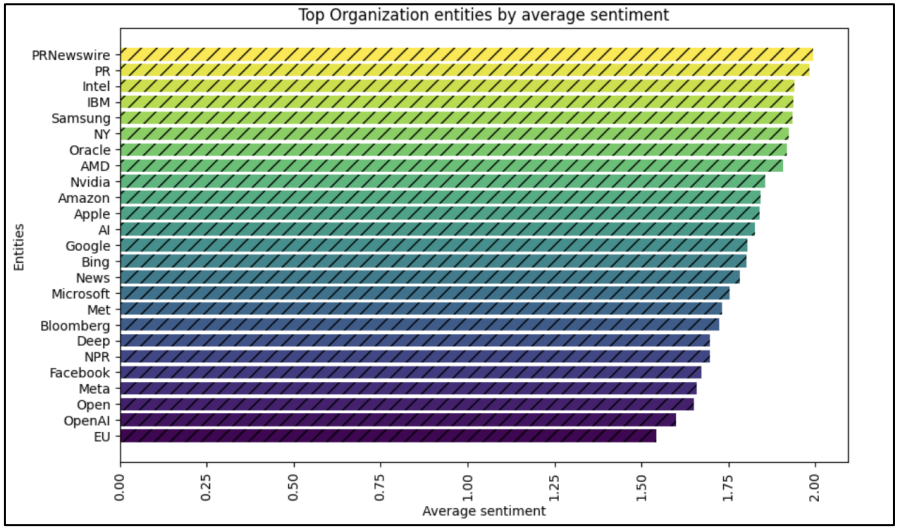
Distribution of sentimental across topics

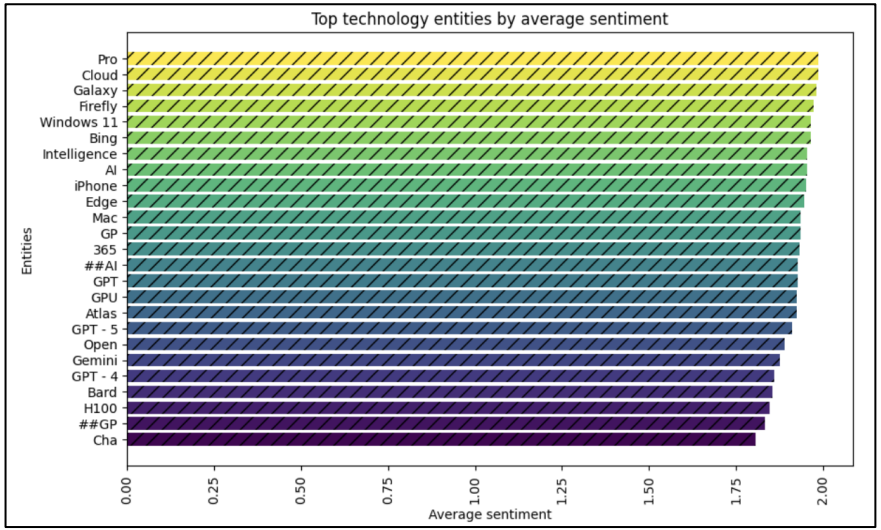


- In overall, it is clear than the model predicts more positive topics than negative or neutral across al topics.
- The perception in the articles tends to more positive compared to others regarding topics: 1(Health & AI), 9 (Enterprise cloud & Data solutions) and 10 Cryptocurrency & Financial Trading. On the other side, writers are more negative about topics such as 2(AI chatbots & workforce) and 7) Tech in the stock market and 4 (Future Digital entertainment & gaming).

Sentimental analysis training model

Distribution of sentimental across Entities



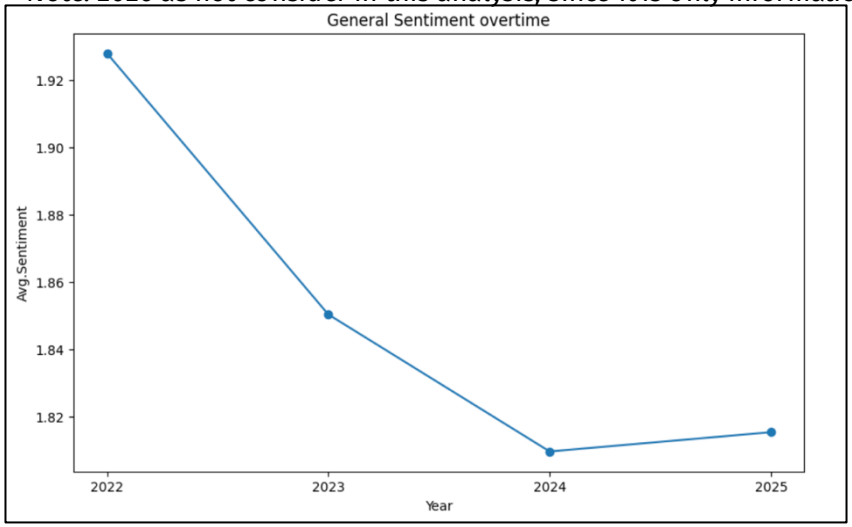


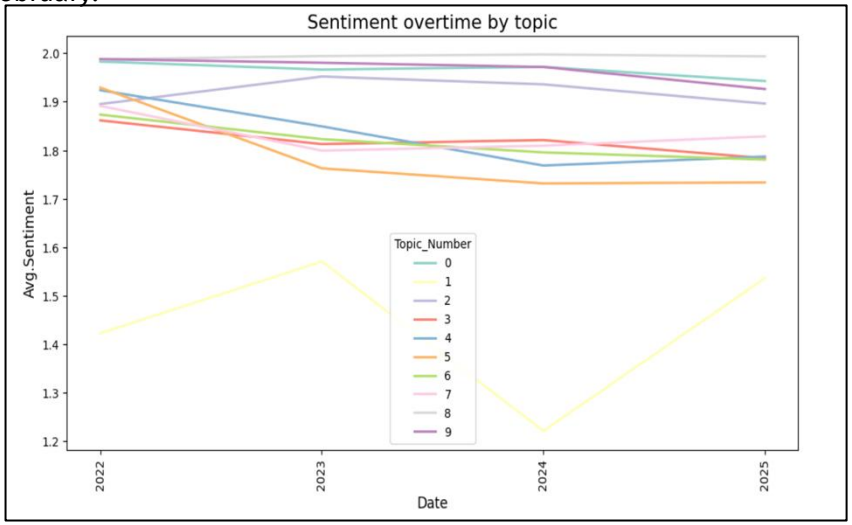
- The tech companies that have less focus on the latest advances of LLMs are viewed with a better sentiment, such as Intel, Oracle, AMD, Amazon, Apple, compared to OpenAI, Meta, Facebook, DeepSeek.
- The technology entities show a similar pattern, people have more positive feelings about traditional tools like Windows, Bing, Edge, Mac, and more negative feelings towards Encoder tools, especially all the products of OpenAI, such as GPT, GPT-4, GPT-5, and Brad, one of the early models of Google.

Sentimental analysis training model

Sentimental Overtime

Note: 2026 as not consider in this analysis, since it is only information until February.

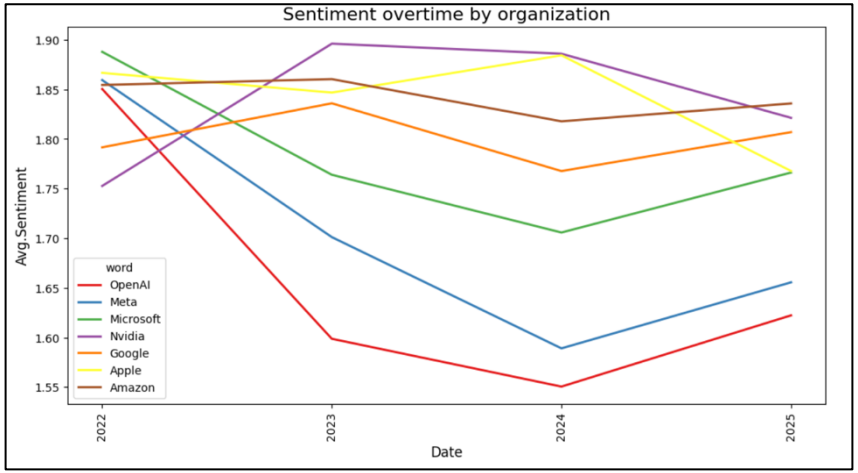


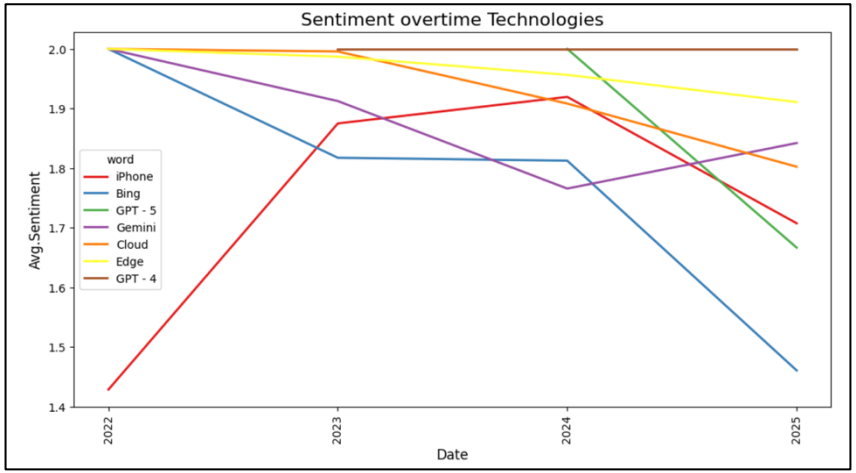


- The general opinion in 2022 about AI was the highest positive feeling on average, but after this period, the articles changed their narrative to more negative or neutral feelings. In the first graph, it is clear to see a considerable decrease until 2024 with 1.8, and a stable trend point in 2025 with 1.82.
- The time series trend sentiment by topic, it is clear that “AI Chatbots & the Workforce (0)” has the most negative feeling compared to the others during the last 4 years, ranging from 1.2 to 1.4. Moreover, ”Generative AI Content Creation(5)” and “Social Media & Tech News(4)” topics have been in constant decline since 2023. Finally, it is worth mentioning that Enterprise cloud & Data solutions(8) have a very constant average positive of around 2.

Sentimental analysis training model

Overtime





- OpenAI and Meta are the companies that have been damaging their public opinion more compared to other tech organizations; on the other hand, Nvidia and Amazon have trended towards a more positive view for writers. In addition, during 2025, Apple recorded the largest decrease in positive feelings compared to the previous year, going from 1.86 to 1.75.
- In the technology's entities sentiment evolution, the biggest change was experienced by Bing tool passing form almost a percent score to 1.5. In addition, the feeling about GPT-5 has decreased significantly in 2025 compared to previous years.

Sentimental analysis training model

Conclusions and recommendations

- According to the topic modeling LDA, the companies in industry such as tech, research, finance, health, social media, cloud solutions, crypto, content creation, and gaming are experiencing the most change with this technology. Therefore, these sectors are and will continue to be affected the most by AI compared to more traditional industries like construction, installation & repair, food & service, and agriculture.
- The big tech companies focused on the development of LLMs are not only changing the tech industry but are also being presented across multiple topics/industries, especially in finance, healthcare, and cloud.
- Traditional tools, such as Bing and Microsoft 365, and devices, like iPhones, Samsungs, and Macs, need to continue evolving to be competitive in this new AI market.
- In general, the articles are more oriented toward positive feelings regarding this AI. However, it is clear that people are not totally happy about these changes, especially in aspects related to “AI Chatbots & the Workforce” and “Generative AI content creation”.
- It is highly recommended that OpenAI, Meta, and Apple start an action plan to reverse the negative trend of neutral or negative feelings people are having about AI in these organizations. Therefore, this will improve the image perception and the successful use cases of AI in several organizations.
- The most positive sentimental use case in “Enterprise Cloud & Data Solutions”, companies that have implemented these technologies are the most satisfied, in contrast with other tools. It will be worth exploring in detail what makes these topics evoke the highest positive feelings, and how that is, since 2022 until 2025 the positive perception has stayed very constant.